



MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of Faculty: Bhanwar Veer Singh

Subject & Subject Code: Engineering Chemistry (1FY2-03)

Semester: I

Session: 2022-23 (Odd Sem.)

Branch: All

Batch: - C and D

Unit No.: 2

Date of Submission: -27.9.22

Lecture No.	Topics Covered	Page No.	Total Page
1	Solid fuels: Coal, Classification of Coal,	1-3	3
2	Proximate and Ultimate analysis of coal & its significance),	4-7	4
3	Gross and Net Calorific value,	8	1
4	Determination of calorific value by bomb calorimeter	9-12	4
5	Metallurgical coke, Carbonization processes; Otto-Hoffmann byproduct oven method	13-16	4
6	Liquid fuels : Advantages of liquid fuels, Mining, Refining and Composition of petroleum	17-18	2
7	Cracking, Synthetic petrol, Reforming, Knocking, octane& cetane number	19-25	7
8	Gaseous fuels; Advantages, manufacturing, composition and Calorific Value of coal& oil gas	26-28	3
9	Determination of calorific value of gaseous fuels by Junker's calorimeter	29-30	2
10	Numerical problems based on determination of calorific value (bomb calorimeter/Junkers calorimeter/Dulong's formula, proximate, ultimate analysis	31-32	2

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

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Lecture No.	Topics Covered	Plagiarism (%)	Unique (%)
1	Solid fuels: Coal, Classification of Coal,	56	44
2	Proximate and Ultimate analysis of coal & its significance),	20	80
3	Gross and Net Calorific value,	14	86
4	Determination of calorific value by bomb calorimeter	20	80
5	Metallurgical coke, Carbonization processes; Otto-Hoffmann byproduct oven method	20	80
6	Liquid fuels : Advantages of liquid fuels, Mining, Refining and Composition of petroleum	30	70
7	Cracking, Synthetic petrol, Reforming, Knocking, octane& cetane number	17	83
8	Gaseous fuels; Advantages, manufacturing, composition and Calorific Value of coal& oil gas	19	81
9	Determination of calorific value of gaseous fuels by Junker's calorimeter	10	90
10	Numerical problems based on determination of calorific value (bomb calorimeter/Junkers calorimeter/Dulong's formula, proximate, ultimate analysis	31	69

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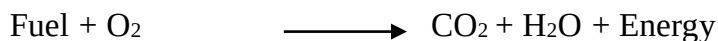
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UNIT-2 ORGANIC FUELS

Introduction

A fuel is any combustible substance that contains carbon (as main constituents), hydrogen, sulphur, etc undergoes combustion in presence of oxygen to gives large amount of energy.



Classification of Fuel

On the basis of occurrence- It is of two types.

- 1) Primary Fuels: It occurs in nature such as wood, coal, petroleum, natural gas.
- 2) Secondary Fuels: It is derived from primary fuels such as coke, gasoline, coal gas.

On the basis of physical state-

General Division	Primary fuels	Secondary Fuels
Solid	Wood, Coal, Peat	Coke, Charcoal, etc.
Liquid	Crude oil	Petrol, Kerosene, Diesel, Tar etc
Gaseous	Natural gases	Producer & Water gas, Coal & Oil gas etc

Characteristics of Good Fuel

- i) Fuel should have high calorific value.
- ii) Must have moderate ignition temperature.
- iii) Fuel should have low moisture content.
- iv) Available in bulk at low cost.
- v) Should not burn spontaneously.
- vi) Fuel should burn efficiently, without releasing hazardous pollutants.
- vii) Handling, storage and transportation should be easy.

SOLID FUEL- COAL, CLASSIFICATION OF COAL

COAL: Coal is a naturally occurring solid fuel, which is formed from plants under high temperature and pressure over millions of years. Coal is highly carbonaceous matter and is mainly composed of C, H, N and O along with some non combustible matter.

CLASSIFICATION OF COAL

(a) Peat

1. Peat is the first stage in the formation of coal.
2. Its calorific value is about 4000-5400 kcal/kg.
3. It is an uneconomical fuel due to its high proportion of (80 -90%) moisture and lower calorific value.
4. It is a brown fibrous mass.

(b) Lignite

1. Lignite is an intermediate stage in the process of coal Formation.
2. Its calorific value is about 6500-7100 kcal/kg
3. Due to the presence of high volatile content, it burns with long smoky flame.

(c) Bituminous Coal- Bituminous coal is further sub-classified on the basis of its carbon content into three types as:

1. Sub- bituminous coal- Calorific value- 7000 kcal/kg
2. Bituminous coal - Calorific value- 8000-8500 kcal/kg
3. Semi-bituminous coal. Calorific value- 8500-8600 kcal/kg

(d) Anthracite

1. Anthracite is the superior grade of coal.
2. Its volatile, moisture and ash contents are very less.
3. Its calorific value is about 8650 kcal/kg.

Unit of heat

- i) **Calorie:** It is the amount of heat required to raise the temperature of 1 gram of water through one degree centigrade.
- ii) **Kilo calorie:** It is the amount of heat required to raise the temperature of 1 Kg of water through one degree centigrade.

iii) **British thermal unit (BTU):** It is the amount of heat required to raise the temperature of 1 pound of water to one degree Fahrenheit.

$$1 \text{ B.T.U.} = 252 \text{ cal} = 0.252 \text{ kcal}$$

$$1 \text{ kcal} = 3.968 \text{ B.T.U.}$$

iv) **Centigrade Heat Unit (CHU):** It is define as the amount of heat required to raise the temperature of 1 pound of water to one degree centigrade.

$$1 \text{ kcal} = 3.968 \text{ B.T.U.} = 2.2 \text{ C.H.U.}$$

References:

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PROXIMATE & ULTIMATE ANALYSIS AND ITS SIGNIFICANCE

Analysis of Coal

The quality of coal can be analyzed by two analysis; proximate and ultimate analysis.

Proximate analysis:

In this analysis moisture, volatile matter, ash and fixed carbon can be determined.

- 1) **Determination of Moisture content:** Powder coal sample taken in a dry weighed crucible. The crucible is heated for about one hour at 105 - 110°C in electric hot air oven. Cool the crucible in desiccator and weight it again. The loss in weight corresponds to the moisture.

$$\% \text{ of moisture} = \text{loss in weight} / \text{weight of coal sample taken} \times 100$$

- 2) **Determination of Volatile matter:** After removing of moisture coal left in crucible kept in an electric furnace at 925±25°C for about 7-10 minutes. It is cooled and weighed. Loss in weight is represented as volatile matter.

$$\% \text{ of volatile matter} = \text{loss in weight due to removal of volatile matter} / \text{weight of coal sample taken} \times 100$$

- 3) **Determination of Ash Contents:** The residual coal sample in the crucible is burn in an open crucible at 700-750°C in a muffle furnace for 30 minutes. It is cooled and weighed. The residue is reported as ash content on % basis.

$$\% \text{ of ash} = \text{weight of ash formed} / \text{weight of sample taken} \times 100$$

- 4) **Fixed Carbon:** It is represented the quantity of carbon in coal that can be burnt. Higher the % of fixed carbon, greater is the calorific value. Thus higher % of fixed carbon is desirable.

$$\% \text{ of fixed carbon} = 100 - [\% \text{ of moisture} + \% \text{ of volatile matter} + \% \text{ Ash}]$$

Importance of Proximate Analysis:

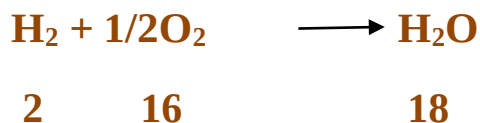
- Moisture reduced the effective calorific value of coal so low moisture content coal is better than high moisture content coal.
- A high volatile matter containing coal burns with a long flame. High smoke and has low calorific value. Hence lesser the volatile matter better is the quality of coal.

- Greater the percentage of fixed carbon, greater its calorific value.
- Ash is a useless, non combustible matter and is reduce the calorific value so less percentage of ash will improve the quality of coal.

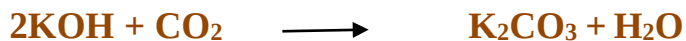
Ultimate analysis:-

Ultimate analysis is involving the measurement of C, H, N, S, and O.

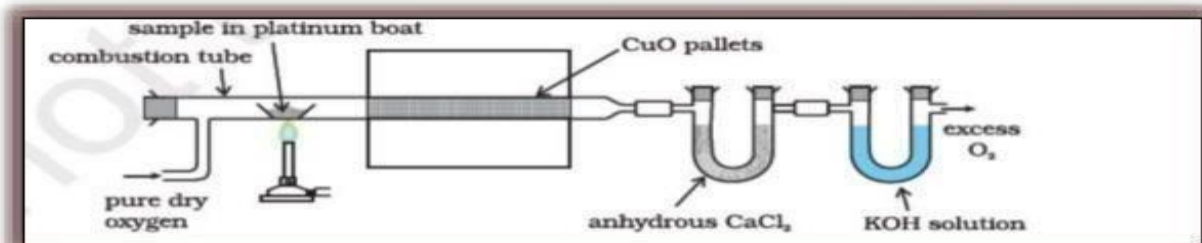
- 1) **Carbon and hydrogen:** In a combustion apparatus, about 1-2 gram of coal sample is burnt in dried oxygen to convert C and H into CO₂ and H₂O respectively.



After this gaseous product of combustion passed to KOH and anhydrous CaCl₂ tubes of known weight. The gaseous products CO₂ and H₂O are absorbed in KOH and CaCl₂ tubes of known weights, respectively. The increase in weights of these (KOH and CaCl₂) are then determined.



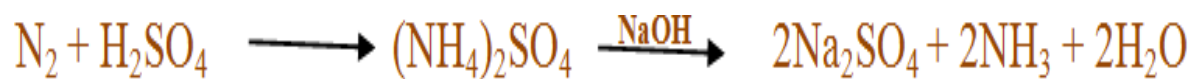
Determination of carbon & hydrogen



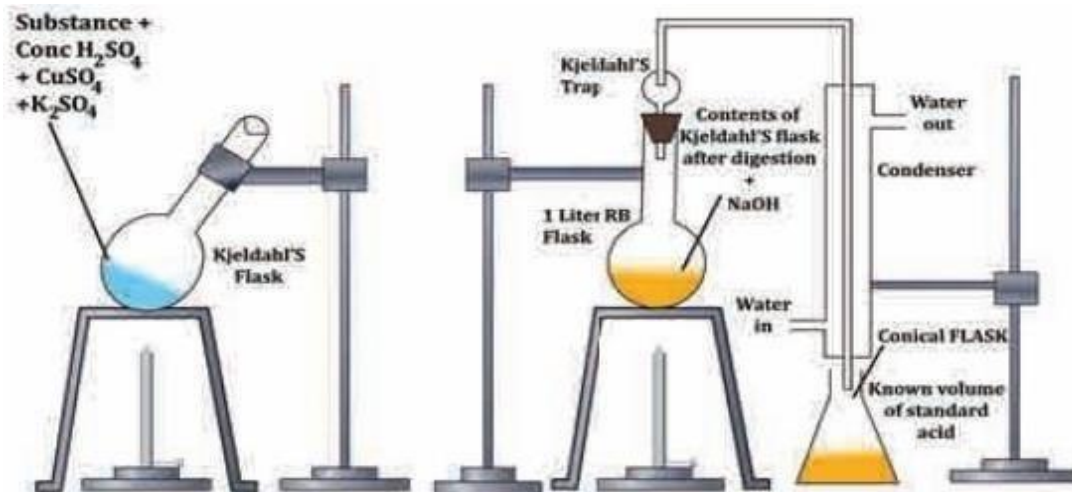
$$\text{Carbon (\%)} = \frac{\text{Increase in weight of KOH} \times 12}{\text{Weight of coal taken} \times 44} \times 100$$

$$\text{Hydrogen (\%)} = \frac{\text{Increase in weight of CaCl}_2 \times 2}{\text{Weight of coal taken} \times 18} \times 100$$

2) Nitrogen: In Kjeldahl's flask add accurately weighed powdered coal and heated with concentrated H_2SO_4 and K_2SO_4 as a catalyst. The solution becomes clear when all the nitrogen is converted into ammonium sulphate then it is treated with excess of NaOH which convert ammonium sulphate into ammonia; the liberated ammonia is distilled over and absorbed in a known volume of standard (N/10) H_2SO_4 solution. From the volume of H_2SO_4 used by liberated ammonia, the percentage of Nitrogen in coal, calculated as follows:



The volume of unused H_2SO_4 is then determined by titrating (back titration) against standard NaOH solution (N/10).



KJELDAHL'S METHOD

The percentage of nitrogen can be determined using the given formula:

$$\text{Percentage of nitrogen} = \frac{1.4 \times \text{volume of acid used} \times \text{Normality of acid (N/10)}}{\text{Weight of coal taken}}$$

3) Sulphur: A known amount of coal sample is burnt in bomb calorimeter in presence of

oxygen. After that, sulphur present in coal is converted into sulphate. The ash obtained from the bomb calorimeter, is extracted with dil. HCl. The washings (acid extracts) are treated with Barium chloride solution and the sulphates are precipitate as Barium sulphate. This precipitate is filtered, washed, dried and heated to obtain constant weight.



$$\text{Percentage of Sulphur} = \frac{\text{Weight of BaSO}_4 \times 32 \times 100}{\text{Weight of coal sample} \times 233}$$

4) **Ash-** The percentage of ash determined as –

$$\% \text{ of ash} = \frac{\text{weight of ash formed}}{\text{weight of sample taken}} \times 100$$

5) **Oxygen:** It is determined by subtracting the sum of total % of carbon, hydrogen, nitrogen, sulphur and ash from 100.

$$\% \text{ of oxygen} = 100 - [\% \text{ carbon} + \% \text{ hydrogen} + \% \text{ nitrogen} + \% \text{ sulphur} + \% \text{ ash}]$$

Significance of Ultimate Analysis:

1. Greater the % of C and H, better is the coal in quality and calorific value.
2. Nitrogen is an inert and incombustible gas and doesn't contribute in calorific value.
3. Sulphur is also contribute in calorific value but it is undesirable because it pollute the atmosphere by SO₂ and SO₃.
4. Oxygen content decreases the calorific value of coal. High oxygen content coals are characterized by high moisture and volatile matter. Thus a good quality coal should have low % of oxygen.

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

GROSS & NET CALORIFIC VALUE

Calorific value: Calorific value of a fuel is defined as “The total quantity of heat liberated, by the complete combustion of unit mass of fuel”.

Types of calorific value

1. **Gross or Higher calorific Value (HCV):** The total amount of heat generated when a unit mass or unit volume of fuel is completely burnt and the product of the combustion are cooled down to room temperature.

$$\text{GCV} = \text{Heat liberated from combustion} + \text{Latent Heat}$$

2. **Net or Lower calorific Value (LCV):** The net heat produced when unit mass or unit volume of the fuel is completely burnt and the products of combustion are allowed to escape”.

$$\text{LCV} = \text{GCV} - \text{Latent heat of water vapour produced}$$

$$\text{LCV} = \text{GCV} - \text{mass of hydrogen} \times 9 \times \text{latent heat of steam (587 kcal/kg)}$$

Latent Heat- Heat liberated as a result of condensation of steam into water is known as latent heat.

References:

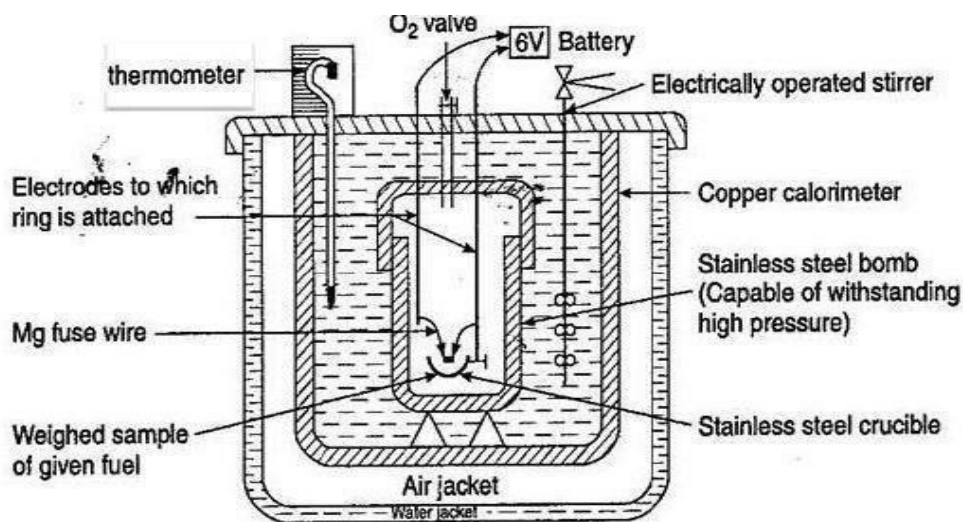
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DETERMINATION OF CALORIFIC VALUE BY BOMB CALORIMETER

Bomb calorimeter

Principle- A known mass of fuel is burnt and the quantity of heat produced is absorbed in water which is measured. Bomb calorimeter is used to determine calorific value of solid and liquid fuels experimentally.

Description: A bomb calorimeter contains a cylindrical bomb made by stainless steel. Combustion takes place in this cylinder. The lid contains two stainless steel electrodes. Oxygen is supplied through oxygen valve for combustion. The electrode is attached with a Mg wire which supports nickel or stainless steel made crucible. The bomb is taken in a copper calorimeter which is surrounded by air and water jacket in order to prevent heat loss by radiation. The copper calorimeter also contains electrically operated stirrer and Beckmann's thermometer (take reading with temperature difference up to 0.01°C).



BOMB CALORIMETER

Working:

- 1) In a crucible, a known amount of the fuel is placed in the nickel or stainless steel crucible.
- 2) A Pt/ Mg wire connected with electrode which is touched with fuel.
- 3) The bomb is filled with oxygen at about 25 atm pressure and covers the bomb tightly with lid.

- 4) The Bomb is placed in copper calorimeter containing a known amount of water. The Electrically operated stirrer is driven and notes the initial temperature of water (t_1) by thermometer.
- 5) Electrodes are connected to a battery to complete the circuit. The fuel sample is burn and heat is liberated which is absorb by water in copper calorimeter, water is continuously stirred and the final temperature (t_2) of water is noted.

Observation

Mass of fuel (solid or liquid) = x g

Mass of water taken = W g

Water equivalent of calorimeter = w g

Initial temperature of water in calorimeter = t_1 K
Final temperature of water in calorimeter = t_2 K

Calculation

Heat absorbed by the water = $W(t_2 - t_1)$ cal

Heat absorbed by the calorimeter = $w(t_2 - t_1)$ cal

Total heat absorbed = $W(t_2 - t_1) + w(t_2 - t_1)$ cal
= $(W + w)(t_2 - t_1)$ cal

Heat liberated by the fuel = $x \times C$

Heat liberated by the fuel = Total heat absorbed
 $x \times C = (W + w)(t_2 - t_1)$ cal

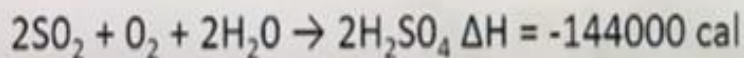
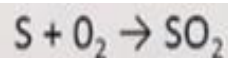
High or gross calorific value (C) = $\frac{(W + w)(t_2 - t_1)}{x}$ cal

LCV = HCV – $\frac{H}{100} \times 9 \times 587$ kcal/kg

Corrections:

To obtain accurate calorific value following corrections are include-

- 1) **Fuse wire correction-** The measured heat also includes the heat given by ignition of fuse wire so this heat is subtracted from total value.
- 2) **Acid correction-** During ignition Sulphur and nitrogen oxidised to sulphuric and nitric acid along with evolution of heat, this heat is subtracted from total value.



- 3) **Cooling correction**- The time taken to cool the hot water in calorimeter from maximum to room temp. is noted. From the rate of cooling and actual time taken for cooling, the cooling correction is added to rise in temp.

$$\text{HCV} = \frac{(W + w) \{(t_2 - t_1) + \text{cooling correction}\} - (\text{Acid correction} + \text{fuse wire correction})}{\text{Mass of the fuel} \times}$$

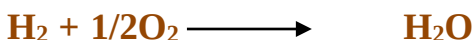
Theoretical calculation of calorific value of solid fuels by Dulong's formula:

The theoretical calculation of calorific value of a fuel can be approximately calculated by Dulong's formula, based on the percentage of the constituents (C, H, O and S) present in the fuel.

Dulong's formula for calorific value from the chemical composition of fuel is:

$$\text{HCV} = 1/100 [8,080 C + 34,500 (H - O/8) + 2,240 S] \text{ kcal/kg}$$

where C, H, O, and S are the percentages of carbon, hydrogen, oxygen and sulphur in the fuel respectively. In this formula, oxygen is assumed to be present in combination with hydrogen as water.



2g 16g 18g

1g 8g 9g

Total mass of hydrogen in fuel – fixed hydrogen

Total mass of hydrogen in fuel – (1/8) mass of oxygen in the fuel

∴ 8 part of oxygen combine with 1 part of hydrogen to form H₂O.

$$\text{Fixed hydrogen} = \frac{\text{Mass of oxygen in the fuel}}{8}$$

Then the amount of hydrogen available for combustion

= total mass of hydrogen in fuel – fixed hydrogen

= [H-O/8]

$$\begin{aligned}\text{LCV} &= [\text{HCV} - 9\text{H}/100 \times 587] \text{ kcal/kg} \\ &= [\text{HCV} - 0.09 \text{ H} \times 587] \text{ kcal/kg}\end{aligned}$$

References:

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Carbonization processes, Metallurgical coke, Otto- Hoffmann byproduct

oven method

METALLURGICAL COKE -The process of preparing coke from coal is known as carbonization of coal. When bituminous coal is heated strongly in the absence of air, the dense, strong, porous mass obtained is called metallurgical coke.

Characteristics of Metallurgical coke-

- i) Purity: - It should contain less percentage of moisture, ash, phosphorus and sulphur.
- ii) Porosity: - It should be porous, so that combustion should be uniform and complete.
- iii) Calorific value: - Coke should possess high calorific value.
- iv) Cost: - Coke should be cheap and easily available.
- v) Combustibility: - Coke should burn easily.
- vi) Reactivity: - The reactivity of coke should not be very high.

Carbonization of coal: When coal is heated strongly in absence of air, it forms a strong, porous, dense, lustrous and coherent mass known as coke. The process of conversion of coal into coke is called carbonization.

There are two types of carbonization:

1. LTC (Low Temperature Carbonization)
2. HTC (High Temperature Carbonization)

1. Low-temperature carbonization: When carbonization is carried out at 500- 700°C is called low temperature carbonization. It cannot be used as metallurgical coke. It is more satisfactory as domestic fuel.

2. High-temperature carbonization: When carbonization of coal carried out at 900-1200°C. is called high temperature carbonization. Hard cokes are obtained by this process which has desired properties strength, purity and hardness. This coke is mainly used for metallurgical purposes.

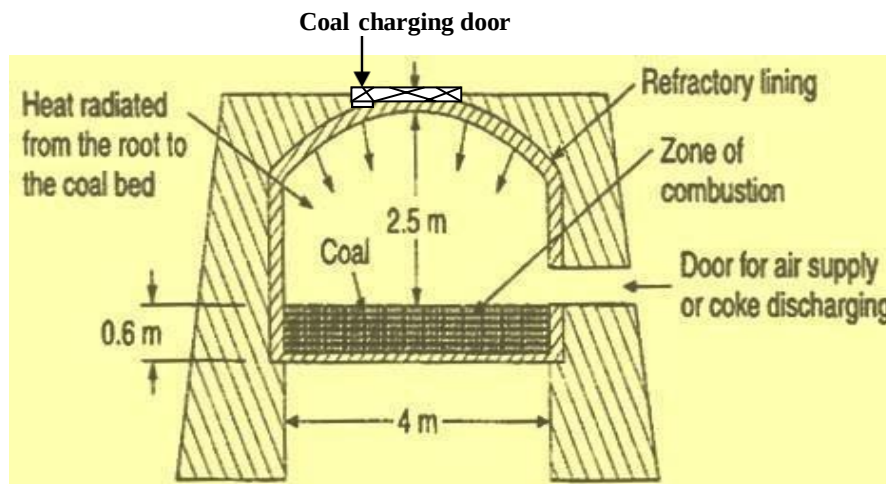
Metallurgical coke: Coke required for metallurgical purpose can be prepared by two methods.

1. **Beehive coke oven method**
2. **Otto Hoffmann oven method**

1.BEEHIVE OVEN METHOD

- 1) Beehive oven is the dome shaped structure called as beehive oven because it resembles beehive in shape.
- 2) It is 4m wide and 2.5 high.
- 3) Roof is provided with an opening for charging the coal. Another opening is present at the side door which works as inlet of air or outlet of coke.

- 4) Coal is introduced from the top and is spread evenly on the floor of oven till the layer of coal becomes 0.6m high.
- 5) Some hot air is supplied initially to ignite the coal for partial combustion.
- 6) It takes 3-4 days in completion of carbonization.
- 7) At the end of carbonization the hot coke is quenched with water and is discharged from side door.
- 8) The yield of coke is 70-80%.



BEEHIVE OVEN METHOD

DEMERITS OF BEEHIVE OVEN METHOD:

- 1) Lower yield of coke due to partial combustion of coal.
- 2) No recovery of byproducts can be done. Byproducts maybe useful but they are allowed to escape.

Manufacture of Metallurgical Coke by “Otto-Hoffmann’s By-Product Oven Process”:-

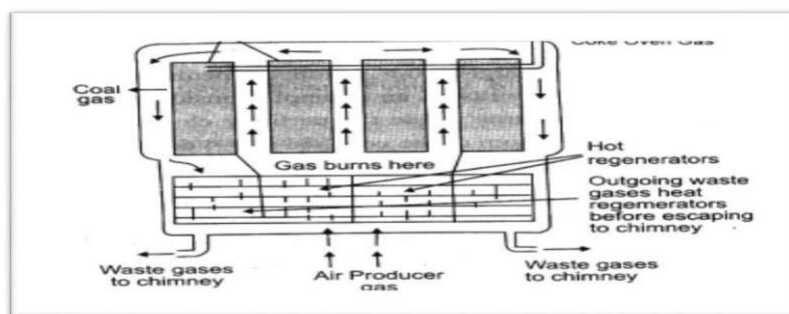
Principle- In order to increase the thermal efficiency of carbonization process and to recover the valuable by-products, Otto-Hoffmann developed a modern by-product coke oven.

Process-

1. It consists of number of narrow silica chambers, each chamber being 10-12 m long 3-4 m height and 0.3 -0.45 m wide.
2. Each chamber is provided with a charging hole at the top, a gas off-take and a refractory lined cast iron door at each end for discharging coke.
3. Coal is introduced into the chamber and the chambers are closed. The ovens are heated to

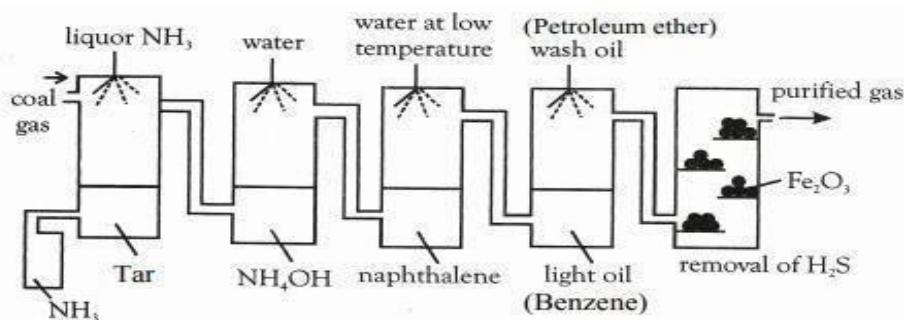
1200°C by burning a gaseous fuel like producer gas and usually employ a regenerative principle to active economical heating.

4. The fuel gases produced during combustion are passed to the regenerators having chequer brick work that takes up the heat and gets heated to about 1000°C. The flow of heating gases is the reversed so that the hot flue gases preheat the other chambers.
5. Carbonization of the charge coal complete in 11 to 18 hours.
6. When coke is formed, It is subsequently quenched by wet quenching (spraying water) or dry quenching (passing inert gas N_2). The coke obtained by dry quenching is drier, clean and stronger than wet quenched coke.



Otto-Hoffmann's By-Product Oven

Recovery of By-products: Coke oven gas is mainly composed of NH_3 , H_2S , Naphthalene, Benzene, tar and moisture.



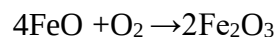
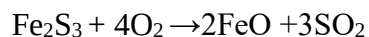
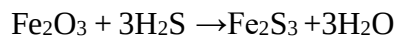
Recovery of By-products by coke oven gas

- 1) **Recovery of Tar:** The gas is first passed through a tower in which liquor ammonia is sprayed. Here dust and tar get collected in a tank below, which is heated by steam coils to recover back ammonia sprayed. The ammonia is used again.
- 2) **Recovery of ammonia:** The gases from the chamber are then passed into a tower in which water is sprayed. Ammonia dissolves in water to form ammonium hydroxide.
- 3) **Recovery of naphthalene:** The gases are allowed to pass into another tower in which water at

very low temperatures sprayed such that the naphthalene is condensed.

4) **Recovery of Benzene:** The gases are then sprayed with petroleum ether to recover benzene.

5) **Recovery of H₂S:** To recover hydrogen sulphide, the gases are allowed to pass through purifying chamber containing moist Fe₂O₃.



References:

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Liquid Fuels, Advantage of liquid fuel, Mining of Petroleum, Refining of Petroleum

ADVANTAGES:-

- Liquid fuels are clean and require less amount of air for complete combustion.
- They possess higher calorific value than solid fuels.
- They are easy to transport through pipes.
- They are used as internal combustion fuels.
- They burn without forming dust, ash, etc.

Disadvantages:-

- Care must be taken during storage.
- Cost is relatively higher than solid fuels.
- They give bad odour.
- Greater risk of fire hazards.

Petroleum or Crude Oil: - The word petroleum is derived from the Latin 'Petra' (rock) and 'Oleum' (oil). It is also known as crude oil or mineral oil. Petroleum is the main source of liquid fuels. Petroleum is a dark greenish-brown viscous liquid found in the earth's crust. The average composition of petroleum shows;

C= 79.5 – 87.1 %

H= 11.5-14.8 %

S= 0.1-3.5 %

O+N= 0.1-0.5 %

Mining of Petroleum:-Petroleum is brought to the surface by drilling holes up to oil bearing area and sinking pipes. Generally, oil is pushed out due to hydrostatic pressure of natural gas. When the pressure of natural gas is not sufficient, then air pressure is applied through lift pumps to force the oil out of the well. Two coaxial pipes are lowered to the oil reservoir, compressed air is forced through the outer pipe, whereby the oil comes out through the inner pipe which is sent to the refineries for further processing.

Refining of Petroleum: - Petroleum obtained from the ground is a mixture of oil and unwanted impurities (water, dirt, sulphur compounds, etc). The process of removal of impurities from crude oil and separation of various fractions with different boiling ranges is called oil refineries. The whole process of refining of petroleum involves the following steps.

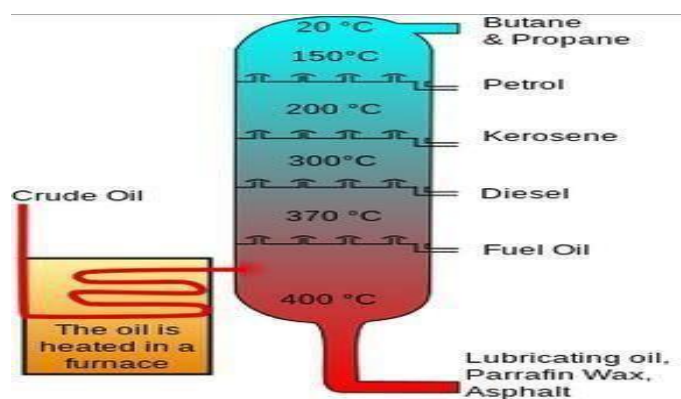
Step-1: Separation of water (Cottrell's process):-

The crude oil is an emulsion of oil and salt water. It is allowed to flow through two highly charged electrodes. The colloidal water droplets coalesce to form large drops which separate out from the oil.

Step-2: Removal of sulphur compounds:-

Sulphur compounds have objectionable properties of pollution. When the crude oil is treated with copper oxide, a reaction occurs with sulphur compounds which results in the formation of copper sulphide ppt which is then removed by filtration.

Step-3: Fractional distillation: - The crude oil is then subjected to about 400°C temp. in an iron retort. All volatile constituents are evaporated. The constituents, tar and asphalt, do not vaporise and they settle at the bottom of the retort. The hot vapours are then passed through a fractionating column. It is a tall cylindrical tower containing a number of horizontal stainless steel trays at a short distance. Each tray is provided with a small chimney, covered with a loose cap. As the vapours go up, they begin to cool and fractional condensation takes place at different levels of column. Higher boiling fractions condense first while lower boiling fractions condense next. Thus the crude oil is fractionated into different fractions depending upon their boiling ranges.

**Fractional distillation of Crude oil****References:**

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

Cracking, Synthetic Petrol, Reforming, Knocking , Octane, Cetane No.-

Cracking - It is defined as the decomposition of high molecular weight hydrocarbon of high boiling point into simple lower molecular weight hydrocarbons of low boiling point. For example,



B. P. = 174°C

B.P. = 36°C

Cracking are two types.

1. Thermal Cracking
2. Catalytic cracking

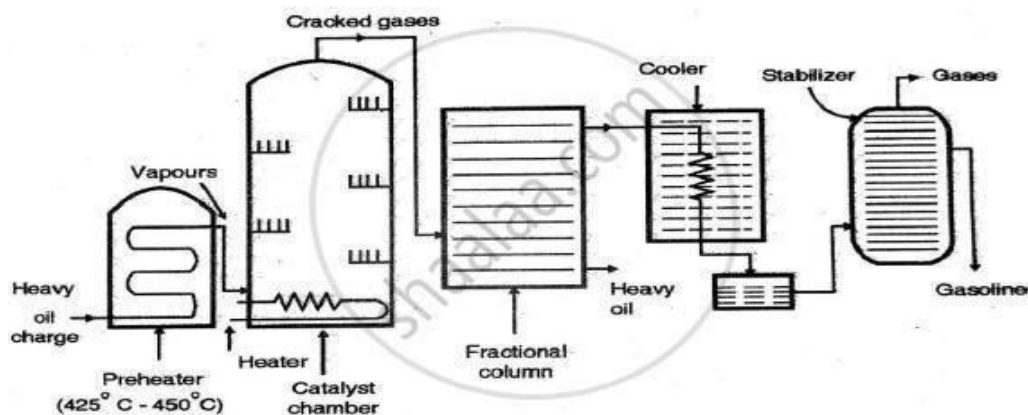
1. Thermal Cracking: This type of cracking is carried out under high temperature and pressure in the absence of any catalyst. It is two types.

- a) **Liquid Phase Thermal Cracking:** In this method, the heavy oil is cracked at a temp. of 475-530°C under high pressure.
- b) **Vapour Phase Thermal Cracking:** In this method oil is vaporized and cracked at about 670-720°C under a low pressure.

2. Catalyst Cracking: In this process the cracking takes place in presence of a catalyst at low temperature and pressure and quality of gasoline so produced is better. Generally the catalyst used is aluminium silicates and alumina. These are two types.

a) Fixed bed catalytic cracking:

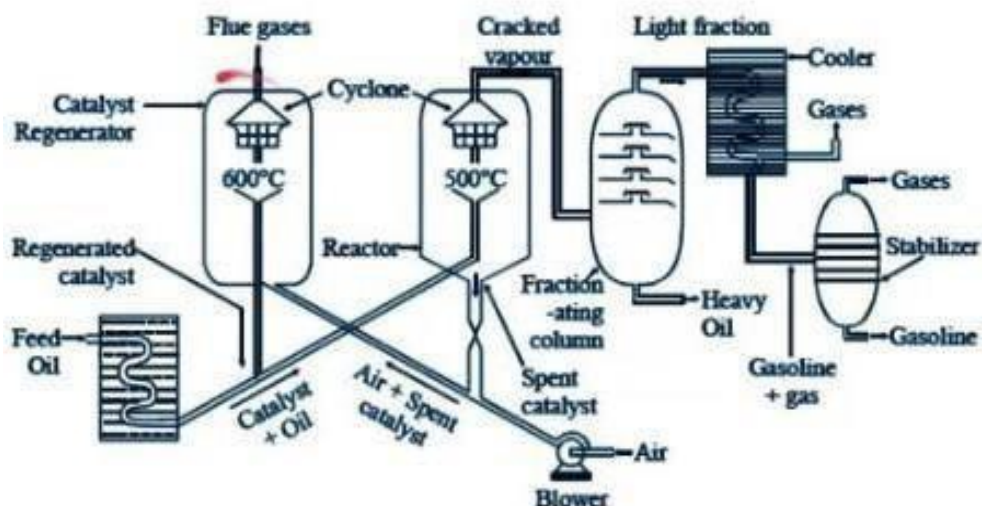
1. The heavy oil is passed through pre-heater at temp. 425-450°C then hot vapours formed.
2. Hot vapours are passed over a fixed bed of catalytic chamber, in this chamber catalyst silica alumina gel present, during the passage through this tower, about 40 % of the charge is converted into gasoline in vapours form and about 2-4 % carbon is formed which one later gets adsorbed on the catalyst bed.
3. Now cracked vapours are passed through a fractionating chamber where heavy oil separated and light fraction is passed through cooler where gasoline condensed along with some gases this is then sent to stabilizer where dissolved gases are removed and pure gasoline is obtained.
4. The catalyst, after 8-10 hours, stops functioning, due to the deposition of black layer of carbon, formed during cracking. This is re-activated by burning off the deposited carbon.



Fixed bed catalytic cracking

Moving bed or Fluidized bed catalyst cracking:

- 1) In moving bed catalytic cracking the feed oil is first passed through a pre heater, which converts oil into vapours.
- 2) The vapours of oil mixed with catalyst are forced up into a large reactor 'bed' in which cracking of the heavier hydrocarbon into lighter molecules.
- 3) Near the top of the reactor, there is a centrifugal separator which allows only the cracked oil vapours to pass on to fractionating column where heavy oil separated, but retains all the catalyst powder in the reactor itself. Light fraction is passed through cooler where gasoline condensed along with some gases this is then sent to stabilizer where dissolved gases are removed and pure gasoline is obtained.
- 4) The catalyst powder gradually becomes heavier, due to coating with carbon, and settles to the bottom, from where it is forced by an air blast to regenerator maintained at 600°C.
- 5) In regenerator, carbon is burnt and the regenerated catalyst then flows through a stand pipe for mixing with fresh batch of incoming cracking oil.



Moving bed or Fluidized bed catalyst cracking

Knocking- Knocking is a sharp metallic sound similar to ratting of a hammer produced in internal combustion engine due to uneven or unsteady or uncontrolled combustion of air gasoline mixture.

In an internal combustion engine when the rate of oxidation becomes so high that the last portion of the fuel-air mixture gets ignited instantaneously, producing an explosive violence known as knocking. A good quality gasoline should resist knocking. The knocking order of different hydrocarbon is -

Straight chain paraffin > branched chain paraffin > olefins > cyclo paraffins > aromatics.

Anti-knocking agents: knocking is decreases by adding some compounds to fuel known as anti-knocking agents.

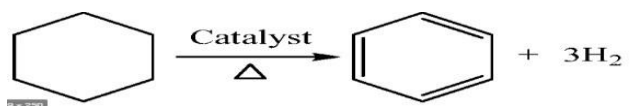
For example: tetra ethyl lead (TEL), methyl cyclopentadienyl magnese carbonyl and iron carbonyl. Tetra ethyl lead (TEL) $\text{Pb}(\text{Et})_4$ known as anti-knock agents. Under combustion, TEL produces Pb and PbO. These particles act as free radical chain inhibitor as they arrest the propagation of the explosive chain reaction responsible for knocking.

Reforming of Gasoline: The process of structural modification by which anti-knocking properties can be improved is called reforming.

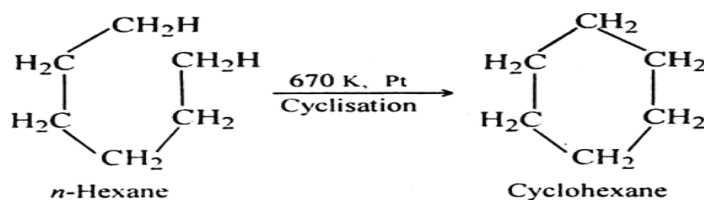
Reforming is the process of rearrangement of straight chains of hydrocarbon in the ring compound.

Example-

Dehydrogenation



Dehydrocyclization



This is two types.

1) Thermal Reforming

2) Catalytic Reforming

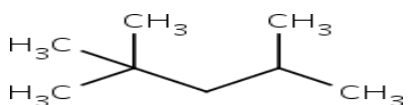
- 1) **Thermal Reforming:** In this reforming is carried out in reactor at 500-600°C under a pressure of 85 atmosphere in the absence of a catalyst. In this process straight chain hydrocarbon converts into branched chain and aromatic hydrocarbon.
- 2) **Catalytic Reforming:** Catalytic reforming is done in the presence of a catalyst (Pt-Al₂O₃) at 460-530°C and 35-50 atmospheric pressure by using either a fixed bed or moving bed cracking.

Octane number: The quality of fuel is determined by its Octane number. Octane number of petrol can be defined as the percentage by volume of iso-octane in the n-heptane and iso-octane mixture in the fuel.

The resistance offered by gasoline to knocking cannot be defined in absolute terms. It has been found that n-heptane knocks very badly and hence its antiknock value (Octane number) has been given ZERO. On the other hand iso-octane (2,2,4- tri methyl pentane) gives very little knocking, so its anti-knock value is given as 100.



n- Heptane
Octane number-0

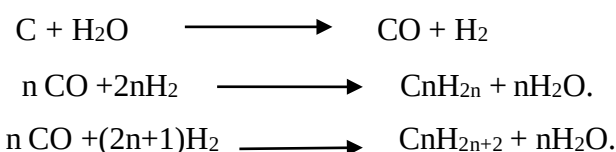


Iso-Octane
Octane number-100

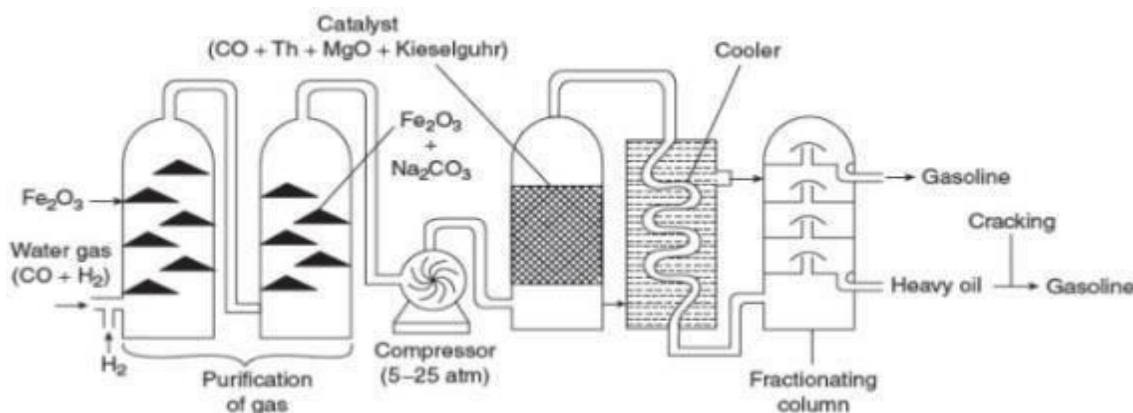
Synthetic Petrol

Synthetic petrol is obtained from coal liquification in which solid coal is converting into liquid. Petrol is generally synthesized by the following two methods.

- ii) **Fischer-Tropsch Process:** - Water gas is a mixture of carbon monoxide and hydrogen ($\text{CO} + \text{H}_2$), produced by passing steam over red hot coke. It is mixed with hydrogen. The gaseous mixture (water gas + hydrogen) is purified by passing over Fe_2O_3 , to remove H_2S , then passing over Fe_2O_3 & Na_2CO_3 mixture to remove organic sulphur compounds. The purified gas is compressed to 5 to 25 atm at $200\text{--}300^\circ\text{C}$ passed through a converter containing catalyst cobalt, thorium, magnesium and kieselguhr. A mixture of saturated and unsaturated hydrocarbons is formed.



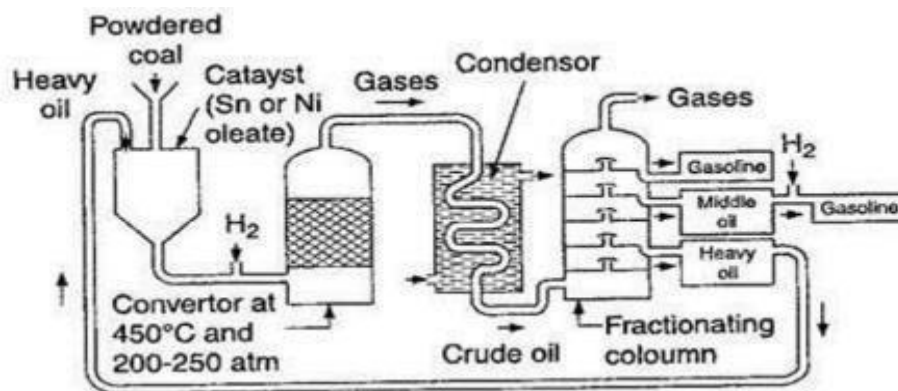
The out coming hot gaseous mixture is cooled in a cooler where a liquid resembling crude oil is obtained which is then subjected to fractionation to yield gasoline and heavy oil. The heavy oil is reused for cracking to get more gasoline.



Fischer-Tropsch Process

- iii) **Bergius Process:** - Coal is powdered and made into a paste with heavy oil and a catalyst is mixed with it. This mixture is sent to a converter maintaining at $400\text{--}500^\circ\text{C}$ and a pressure of 200-250 atm, where it meets the hydrogen. Hydrogenation takes place to form higher hydrocarbons. These are further decomposed to produce lower hydrocarbons. When these lower hydrocarbons are passed through a cooler, a liquid resembling crude oil is obtained. This oil on fractional distillation gives gasoline, middle oil and heavy oil. The middle oil is subjected to hydrogenation in presence of a

catalyst to produce more gasoline. The heavy oil is recycled to make a paste with fresh coal powder.

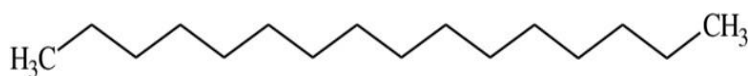


Bergius Process

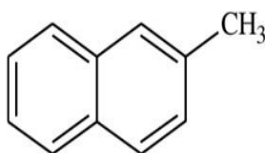
Cetane number: Cetane number of diesel oil may be defined as the percentage of cetane (n-hexadecane) in a mixture of cetane and α -methyl naphthalene, which will have the identical ignition characteristics as the fuel under test, under the same set of conditions. It measures the ease with which the fuel (diesel fuel) will ignite under compression. So higher the cetane number better is the ignition property of the fuel.

The order with decreasing cetane number is as follows n-alkanes > naphthalene > alkenes > branched alkanes > aromatics.

The combustion of fuel in a diesel engine is not instantaneous but there is some lag. The interval between the start of fuel injection and its ignition is called ignition delay or induction lag. Cetane has very short ignition delay compared to any diesel fuel whereas α -methyl naphthalene has a longer ignition delay



n-Hexadecane (cetane), cetane number = 100



1-Methylnaphthalene, cetane number = 0

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram

Gaseous Fuels, Advantages, manufacturing, composition and calorific value of coal gas& oil gas

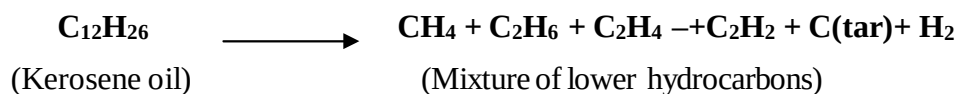
Advantages:-

- They possess high calorific value.
- They easily transport through pipe lines.
- They require less amount of air for complete combustion.
- They burn without smoke and produce no ash.

Disadvantages:-

- They are highly inflammable.
- They require very large storage tanks.
- They are costly than solid & liquid fuels.

Oil gas: Oil gas burns with smoky flame. It is mixture of hydrogen and lower hydrocarbons. It is obtained by thermal cracking of kerosene oil.



Composition of oil gas

$\text{CH}_4 = 25\text{-}30\%$

$\text{CO} = 10\text{-}15\%$,

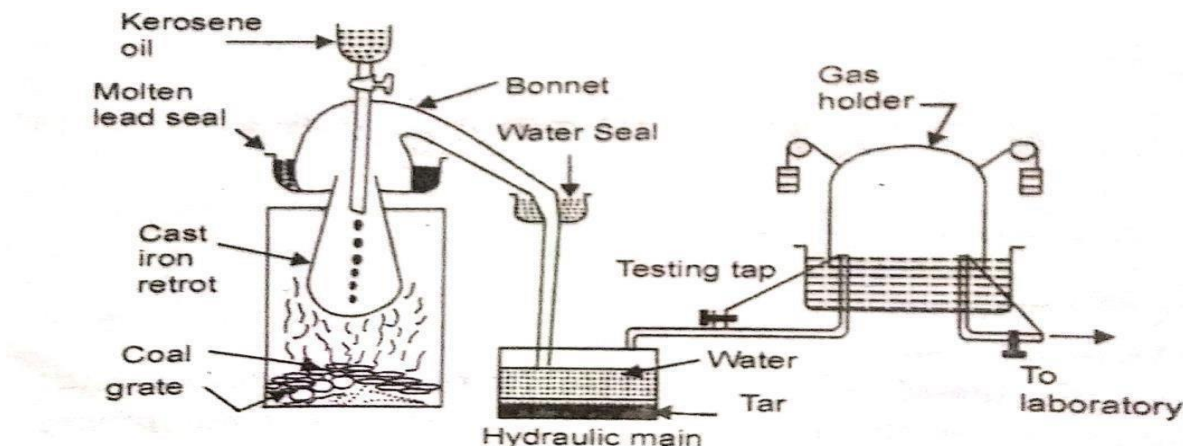
$\text{CO}_2 = 3\%$

$\text{H}_2 = 50\text{-}55\%$

Manufacturing Process:

- Kerosene oil is fed in to a strong iron retort which is enclosed in a coal fired furnace.
- At the mouth of retort, a bonnet is fixed through a molten lead seal.
- The bonnet is then attached a hydraulic main from where a pipe leads to the gas holder.
- The Kerosene oil which is fed in to the red hot retort immediately gets cracked in to a number of lower gaseous hydrocarbons.
- Pure gas can be collected over the gas holder.
- The proper cracking of the oil is assessed from the colour of the gas the produced which should have a golden colour.
- Finally, the gas mixed with air and supplied to the place of burning.

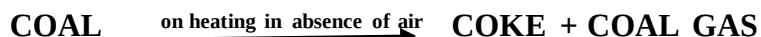
Properties: Oil gas burns with smoky flame and Calorific values about $4500\text{--}5400\text{ k cal/m}^3$.



MANUFACTURE OF OIL GAS

Coal gas: Coal gas is obtained by heating of coal in absence of air by carbonization process.

Coal gas is a mixture of CH_4 (32%) and H_2 (40%).



Composition of coal-

$\text{CH}_4 = 32\%$

$\text{CO} = 7\%$

$\text{CO}_2 = 1\%$

$\text{C}_2\text{H}_2 = 3\%$

$\text{H}_2 = 40\%$

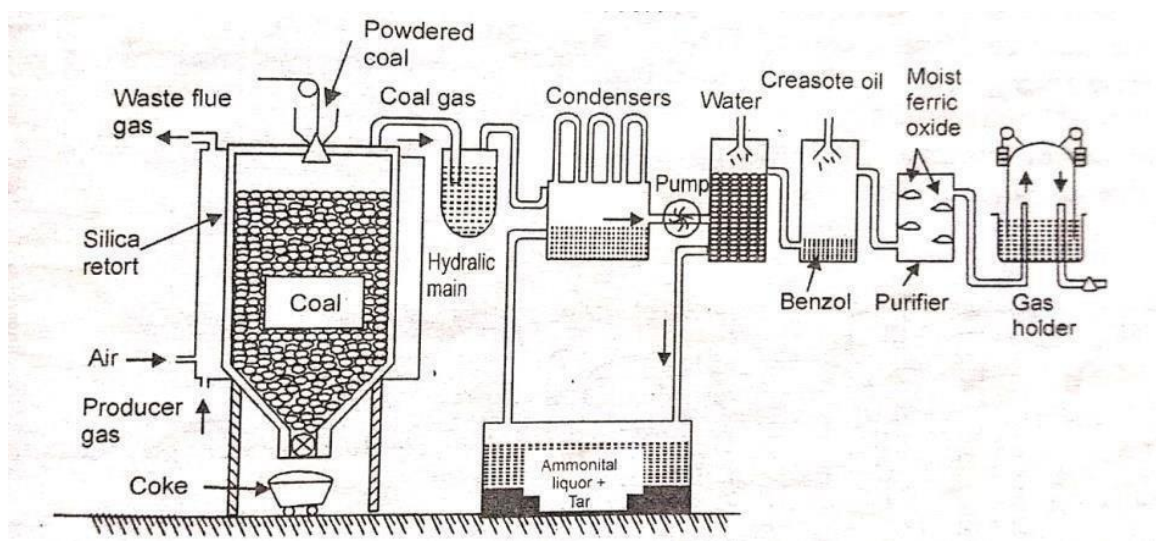
$\text{C}_2\text{H}_4 = 3\%$ and other hydrocarbon = 4 %

Manufacturing Process:

Coal is introduced into the silica chamber and the chambers are closed. The ovens are heated to 1200°C by burning a gaseous fuel like producer gas then by carbonization process coke and coal gas produced that contains large amount of impurities like ammonia, tar organic nitrogen and sulphur compounds. These are removed by passing in to hydraulic main after that scrubbing with creosote oil, which dissolve benzol. Now gas passed over moist ferric oxide for removal of H_2S .

Properties of coal gas:

- Coal gas is colourless.
- It is lighter than air and burns with non smoky flame.
- It is used as domestic fuel for heating.
- Calorific value is about 4900 kcal/m^3 .



MANUFACTURE OF COAL GAS

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C. Kuriacose & J. Rajaram

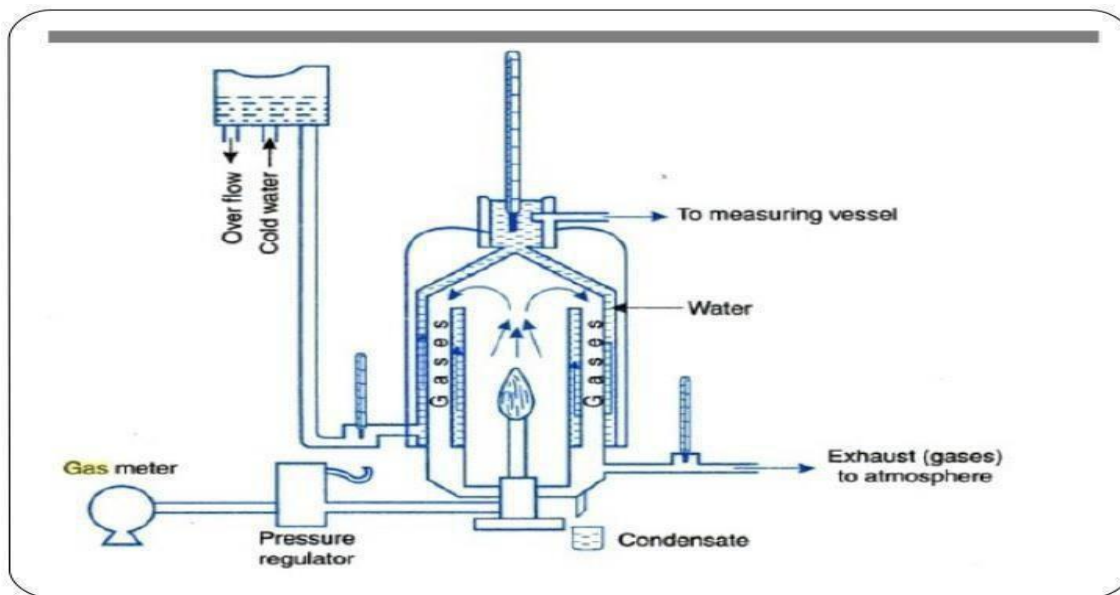
Determination of calorific value of gaseous fuels by Junker's calorimeter

Principle- A known volume of gas fuel is burnt and the quantity of heat produced is absorbed in water which is measured. Junker calorimeter is used to determine calorific value of gases fuels experimentally.

Construction

1. Junkers gas calorimeter consists of a combustion chamber surrounded by water jacket through this water jacket a number of tubes passes.
2. A gas pipe line is connected with a burner kept in combustion chamber.
3. A gas meter and pressure regulator is provided in a gas pipe line.
4. Thermometers are used to measure the temperature of water at inlet, outlet and flue gas.
5. Condensate from the gases is collected in condensate pot.

Junker's calorimeter



Working:

2. The gas whose calorific value is to be measured is supplied through a pipeline to the gas burner where it is burnt.
3. The flow rate of gas is measured by a gas meter. The pressure of gas is measured by a manometer attached to the pressure regulator.
4. The heat produced by combustion of gas is absorbed by the cold water passes through tubes present in water jacket.

5. The gases are cooled up to room temperature as far as possible, so that entire heat released from the combustion may be absorbed by circulating water. The temperature of cooling water at inlet, outlet and exit gas temperature is measured.

Observations:

Volume of gas fuel burnt at given temp. and pressure = V

Volume of water passes through the calorimeter = W (kg)

The mass of water (steam) condensed in the outlet water = m (kg)

Temp. of inlet water = t_1

Temp. of outlet water = t_2

Higher calorific value of fuel = C

Calculation: heat absorbed by circulating water = $W(t_2 - t_1)$

Heat produced by the combustion of fuel = VC

Heat produced by the combustion of fuel = Heat absorbed by circulating water

$$VC = W(t_2 - t_1)$$

$$C(\text{HCV}) = W(t_2 - t_1)/V$$

Lower calorific value = Higher calorific value – $(m/V \times 587)$ Kcal/m³

$$\text{L.C.V} = [C - m/V \times 587] \text{ Kcal/m}^3$$

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C. Kuriacose & J Rajaram

Numerical

Q. A sample of coal contains C = 92 %, H= 5% and ash=3%. The following data were obtained when the above coal tested in a bomb calorimeter experiment. wt. of coal burnt = 0.95gm, wt. of water =700gm, water equivalent calorimeter = 2000gm, Increase in temperature = 2.48° C, Acid correction = 60 cal, cooling correction = 0.02 °C, Fuse wire correction = 10 cal. calculate net and gross calorific value.

Ans. C = 92%, H=5% , Ash = 3% .

Weight of coal burnt = 0.95 gm

Weight of water = 700 gm (W)

Water equivalent = 2000 gm

(w) Increase in temp = 2.48° C

Acid correction = 60 cal Cooling correction = 0.02° C Fuse wire correction = 10 cal Acc. to formula,

$$C = \frac{(W + w) \{ (t_2 - t_1) + \text{cooling correction} \} - (\text{Acid correction} + \text{fuse wire correction})}{\text{Mass of the fuel}}$$

$$C = \frac{(700 + 2000) (2.48 + 0.02) - (60 + 10)}{0.95}$$

$$0.95$$

$$= (2700) (2.50) - (70) / 0.95$$

$$= 6680 / 0.95$$

$$= 7031.57 \text{ cal/g}$$

$$\text{LCV} = \text{HCV} - (0.09 \times H \times 587)$$

$$= 7031.57 - (0.09 \times 5 \times 587)$$

$$= 7031.57 - 264.15$$

$$= 6767.42 \text{ cal/g}$$

Q. A sample of coal has following composition by mass C = 70 %, O = 8 %, H = 10 %, N = 3 %, S = 2%, Ash = 7 %. Calculate H.C.V. and L.C.V. using Dulong formula.

Ans. HCV = $\frac{1}{100} [8080 C + 34500 (H - O/8) + 2240 S]$

$$= \frac{1}{100} [8080 \times 70 + 34500 (10 - 8/8) + 2240 \times 2]$$

$$= \frac{1}{100} [565600 + 34500 (10 - 1) + 4480]$$

$$= \frac{1}{100} [565600 + 310500 + 4480] = \frac{1}{100} [880580]$$

$$= 8805.80 \text{ kcal/kg}$$

$$\text{LCV} = [\text{HCV} - 9/100 H \times 587] \text{ kcal/kg}$$

$$= [8805.80 - 9/100 \times 587] = [8805.80 - 528.30]$$

$$= 8277.80 \text{ kcal/kg}$$

References:

1. Engineering Chemistry, by S. Rattan
2. Engineering Chemistry, Wiley India
3. Chemistry in Engineering & Technology (Vol. I & II) by J.C.Kuriacose & J Rajaram